SQL — THE DIGITAL LIBRARY

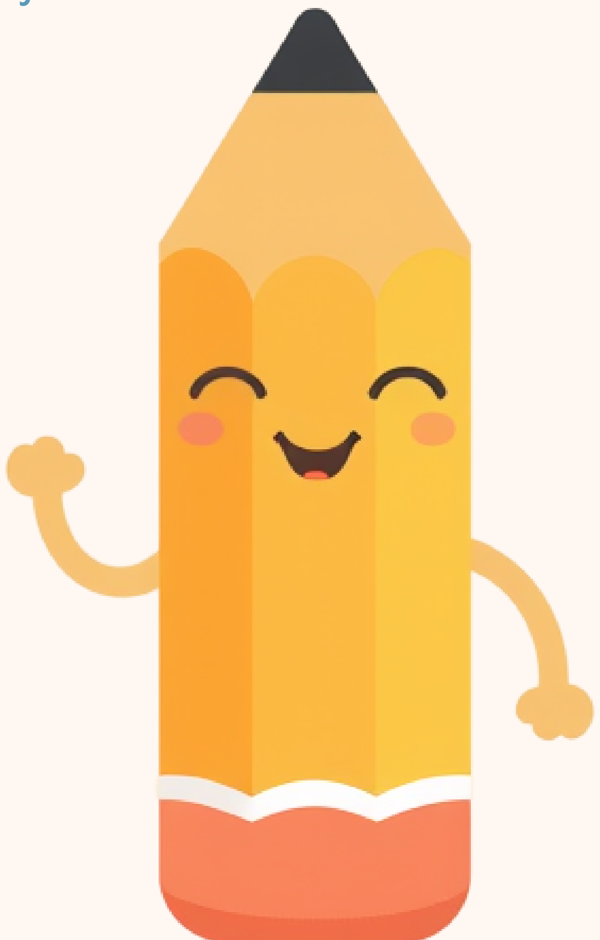
THE DATA EXPLOSION

WHY CAN'T WE USE ONE GIANT NOTEBOOK?

Every day, Indians create billions of pieces of data! From school attendance to cricket scores, library books to train tickets — there's just too much information to keep in one notebook. We need smart storage systems to organize it all!

WHAT IS A RELATIONAL DATABASE?

A Relational Database is Like a Digital Library! It's an organized collection of related tables that store information neatly, just like bookshelves in your school library keep books organized by subject.



THE LIBRARY ANALOGY 📚

📚 Tables = Shelves (organize data)
📖 Rows = Books (each record)
✏️ Columns = Book Info (title, author, pages)

TABLES, ROWS & COLUMNS EXPLAINED

UNDERSTANDING THE STRUCTURE 📊

Think of a table like your class attendance register! Rows go across (each student), columns go down (Roll No 🔢, Name 👤, Date 📅). Each little box (cell) holds just one piece of information.

REMEMBER THIS! 💡

Rows = Each student's record
Columns = Types of information
Cell = One single piece of data

No data to display

Source: AI-generated data. Replace or verify before use.

OUR SCHOOL REGISTER IS A TABLE! 🏫

WHY DO WE NEED A PRIMARY KEY? 🔑

Imagine your class has three students named 'Rahul'! How do we tell them apart? That's where a Primary Key helps! Every student has a unique Aadhaar Number or Admission Number that belongs only to them. 🔢

PRIMARY KEY KEEPS DATA UNIQUE

Think of it like the token number at Tirupati Laddu counter—only ONE person gets each token! No confusion, no mix-ups! 🎫

Primary Key = Your data's special ID card!

PRIMARY KEY IN ACTION

🔢 ADMISSION NO IS THE PRIMARY KEY!

Notice how three students are named Rahul, but each has a unique Admission Number. This prevents confusion!

SAMPLE STUDENT ATTENDANCE TABLE

🔢 Admission No | 👤 Name | 🏫 Class | 📅 Date
A101 | Rahul Sharma | 6A | 15-Jan
A102 | Priya Patel | 6B | 15-Jan
A103 | Rahul Verma | 6A | 15-Jan
A104 | Rahul Singh | 6C | 15-Jan

STUDENT TABLE

Student ID 🔢	Name 👤	Class 🏫
001	Priya	6A
002	Arjun	6B
003	Meera	6A

The Student ID connects to other tables!

LIBRARY BOOKS TABLE

Book ID 📖	Title 📚	Student ID 🔢
B101	Panchatantra	001
B102	Malgudi Days	003
B103	Science Fun	001

EVERYDAY INDIAN EXAMPLES OF DATABASES

Databases are everywhere around us! Here are some examples you might use in your daily life. Each one stores information in tables with rows and columns.

ACTIVITY TIME! DESIGN A CRICKET TEAM TABLE 🏏

Be a Database Architect! Design your own Cricket Team table with these columns:

🔢 Player ID (Primary Key)

👤 Name

🎯 Specialty (Batsman/Bowler/All-rounder)

🏃‍♂️ Runs Scored

Think: What makes each player unique? That's your Primary Key!

SAMPLE CRICKET TEAM TABLE 🏏

🔢 PLAYER ID (PRIMARY KEY)

P001 - Virat (Batsman)
P002 - Rohit (Opener)
P003 - Jasprit (Bowler)

👤 NAME & 🎯 SPECIALTY

P001: 1500 runs
P002: 1200 runs
P003: 50 runs
Notice how IDs keep data organized!

🏃 RUNS SCORED

Player ID is the Primary Key — each player has a unique ID, even if names repeat!

RECAP: WHY ARE DATABASES IMPORTANT?

📊 ORGANIZE DATA EASILY

Databases help us store and manage lots of information in neat tables—like having a super-organized digital notebook!

🔑 PRIMARY KEYS KEEP DATA UNIQUE

No more confusion! Primary Keys ensure every record is unique—like your Aadhaar or Admission Number.

🔗 CONNECT & POWER APPS

Relational databases link tables together and power apps you love—Zomato, Amazon, Disney+ Hotstar!

REAL WORLD: DATABASES AROUND YOU 🌏

Databases power the apps and websites you use every day! From ordering food on Zomato to booking train tickets on IRCTC, databases keep everything organized and running smoothly.

THANK YOU!

Thank you for learning about databases! Any questions?